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The Debate Over Nepal's 300% Vehicle Tax Misses the Real Issue

Nepal's EV boom is quietly testing the country's border tax model

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A viral debate over Nepal's steep import taxes, sparked by comments from Swarnim Wagle, one of Nepal's leading economist-politicians, has prompted a closer look at numbers. While his remarks were largely interpreted as a call to slash the country's roughly 300 percent vehicle tax regime, the data suggests a more systemic issue he hinted at may be real: Nepal's fiscal stability remains dangerously over-reliant on border taxation.

The current EV boom, meanwhile, does not appear to represent true market expansion. Much of the shift reflects existing car owners replacing older petrol vehicles with electric ones, often moving from low-priced ICE cars to mid-range EVs. If this trend persists without comprehensive tax reform, the transition will inevitably erode a revenue model that has, for decades, been built almost exclusively on taxing vehicles at the point of entry.

While Nepal's vehicle tax system rarely sets social media alight, a recent firestorm surrounding Swarnim Wagle, the economist-politician widely seen as a potential Finance Minister in Nepal's first post-Gen-Z parliament, has exposed the raw nerves of the nation's political economy.

The viral debate was sparked by claims that [Wagle proposed a total repeal of Nepal's notorious vehicle taxes](#), which often comes around 300% when customs, excise, and VAT and RDF (Road Development Fees) converge. Wagle was quick to issue a [disclaimer](#), clarifying that his remarks were read out of context and that his argument was never for



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a full 300% withdrawal, but for the rethinking of a system that many economists dismiss as excessively complex and distortionary.

Still, the controversy raised a question worth asking: If these taxes are so politically sensitive and extraordinarily high, how important are they really to Nepal's survival? The real story isn't a simple binary between overtaxing Internal Combustion Engines (ICE) or subsidizing electric vehicles.

Instead, the EV surge is acting as a diagnostic tool, revealing a state dangerously addicted to border-point revenue. As Nepal transitions toward a greener fleet, this market shift signals the exhaustion of a rentier tax model that has reached its logical limit, forcing a long-overdue reckoning with a system the state can no longer afford to ignore.

The 'Fiscal Cliff' Myth

In the conventional economic policy discourse in Nepal, the 300% vehicle tariff is often canonized as a fiscal necessity, a vital bulwark against a national fiscal cliff, and as a check against congestion. The logic was simple: since our road expansion hasn't caught up with urban demand, the state used the tax code as a blunt instrument to keep cars off the street. But this volume check has become a fiscal crutch that masks our failure to build a modern transport network.

Yet, the math suggests that the fiscal cliff is a ghost story. While Nepal's tax revenues are undeniably reliant on imports (42% of tax revenue), vehicles and parts (HS 87) account for a surprisingly contained 7% of total tax revenue.

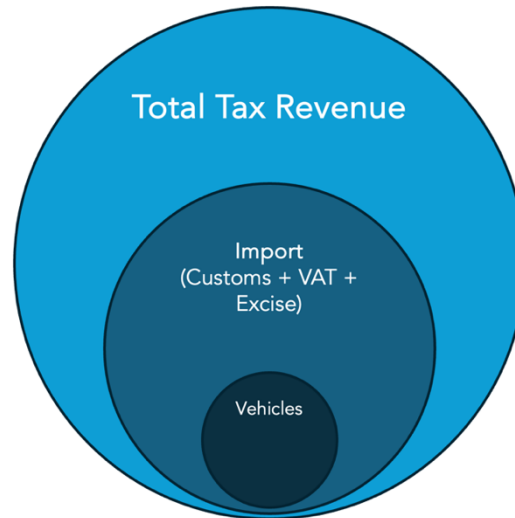


Figure 1: Proportional Representation of Tax Revenue. Source: Ministry of Finance (Total Tax and Import Revenue), Department of Customs (Vehicles), for Fiscal Year 24-25

Our modeling reveals that even a 50% reduction in the effective tax rate for Internal Combustion Engines (ICE) results in a total revenue dip of up to 0.7% (**Figure 2**). In any other sector, this would be dismissed as a rounding error. This budgetary insurance exists because of the Composition Effect: with EVs now constituting more than 70% of the market, the ICE tax base is now "too small to fail" the treasury.

Reduction of Effective Tax Rate on ICE	Impact on Vehicle Tax Revenue			Impact on Overall Tax Revenue		
	Low Case	Expected	High Case	Low Case	Expected	High Case
10%	-0.3%	-0.7%	-1.0%	0.0%	0.0%	-0.1%
20%	-1.0%	-1.8%	-2.2%	-0.1%	-0.1%	-0.2%
30%	-2.1%	-3.0%	-3.6%	-0.1%	-0.2%	-0.3%
40%	-3.9%	-4.7%	-5.3%	-0.3%	-0.3%	-0.4%
50%	-3.5%	-6.9%	-10.6%	-0.2%	-0.5%	-0.7%

Figure 2: Modeling Results

To a casual observer, reducing taxes on Internal Combustion Engine (ICE) vehicles by 50% appears regressive, a retreat from our "Green Revolution." However, a deeper



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econometric analysis reveals the opposite. The current 300% border tax is merely a proxy for the volume check on Nepal's road network. By taxing the entry (ownership) instead of the usage (congestion or road-time), the government has chosen a path of administrative ease over economic efficiency. We are using the tax code to mask a failure in urban planning and smart infrastructure management.

These are the conditions that Swarnim Wagle must now navigate as he prepares to take on the seat of Finance Minister, stepping directly into a high-stakes debate that he himself started. With a 0.7% revenue risk serving as his fiscal safety net, Wagle now has the historic mandate to turn a viral debate into a masterstroke of experimenting some tax restructuring. The question is no longer about the rate, but the intent: is the state using fiscal inertia to mask a logistics failure? It is time to shift from a 'gatekeeper' tax model that punishes entry to an 'enabler' model that manages usage? Nepal can afford to fix its border; it can no longer afford to ignore the math.

Data

Sources and Preparation

Our analysis is built on monthly trade figures sourced directly from the [Customs Department's](#) Foreign Trade Statistics Dataset. To ensure the integrity of the modeled outcomes, we have applied two critical filters to the dataset:

1. **Segment Isolation:** The study is strictly confined to consumer passenger vehicles (HS Code 8703), with special attention paid toward excluding commercial vehicles, buses, ambulances, and two/three-wheelers. These segments operate under distinct tax regimes and exhibit unique demand elasticities that would otherwise introduce noise into the consumer model. We also account for annual changes in descriptions, wherein the HS code from one year refers to a different capacity class in another.
2. **Temporal Integrity:** The usable sample comprises 61 observations beginning in FY 2077-78 (2021). Data prior to this period was discarded to avoid structural bias; Reported data was flawed and distorted due to COVID, and early EV volumes were statistically too thin to provide a reliable basis for market-share estimation.

To prevent month-to-month anomalies from inflating standard errors, we have normalized the imports dataset using a rolling 3-month smoothing technique. By neutralizing the 'sharp spikes' inherent to fiscal boundaries and inventory balancing, we preserve the integrity of the underlying trends. The result is a model that isolates the impact of regulatory change.

We also pull in historical tax regimes from the Ministry of Finance [website](#).

Policy Events in the Sample Period

Our model accounts for three seismic policy pivots that occurred within the sample window. We have isolated these events to ensure that our findings reflect underlying market elasticity rather than temporary regulatory noise:

1. **The 'Carrots' ([Fiscal Liberalization/Paudel Reversal](#)):** The fiscal liberalization was a landmark shift in Nepal's automotive tax policy led by Bishnu Prasad Paudel during his tenure as Finance Minister in 2021. It was primarily a response to the controversial "Khatiwada Budget" of 2020: former Finance Minister Yuba Raj Khatiwada had significantly increased excise and customs duties on electric vehicles (EVs). This move was widely criticized for pricing clean-energy cars higher than their petrol or diesel counterparts, directly contradicting Nepal's green energy goals. The Paudel Reversal effectively broke the market's inertia. By slashing duties, the government created an artificial price advantage that forced consumers to reconsider their long-standing ICE loyalties. This turned out to be a market-shaping event that recalibrated the baseline for "value" in the automotive sector and laid the groundwork for EVs to eventually command over 70% of the passenger vehicle market we see today.
2. **The 'Sticks' ([The ICE Import Ban \(April–December 2022\)](#)):** Facing a balance-of-payments crisis and rapidly depleting foreign exchange reserves, Nepal imposed a near-total ban on ICE vehicle imports from April through December 2022. EVs, which are imported primarily from China under more favorable trade terms, were exempt. We model this as a dummy variable in both equations.
3. **The Parity Precipice:** The modest tax increases in FY 2081 (2024/25) and the decision to hold those rates steady for FY 2082/83 (2025/26) represent a fragile

equilibrium. While EVs still dominate, the recent decline in EV imports in early 2026 highlights the market's extreme sensitivity to border disruptions and even the rumor of further tax hikes.

Modeling and Results

EV Shares – The Wallet Drives the Wheel

In **Model 1 — EV Share (Logit Model)**, we model the log-odds of EV market share as a function of the tax ratio (EV taxes relative to ICE taxes), the price ratio (EV import value relative to ICE import value), a secular time trend, quarterly seasonality, and the import ban dummy. All continuous regressors are standardized (z-scored) prior to estimation to reduce multicollinearity. We also estimate a ridge regression (L2 penalty, $\alpha = 1.0$) to further regularize coefficient estimates. Our regression equation is:

$$\text{logit}(EV Share_t) = \alpha + \beta_1 Trend_t + \beta_2 \log(TaxRatio_t) + \beta_3 \log(PriceRatio_t) + \gamma_s + \delta ImportBan_t + \varepsilon_t$$

In this formulation, γ captures seasonality fixed effects.¹ This model achieves an adjusted R^2 of 0.836 with HC1 robust standard errors, and a low condition number.

¹ Note that while Lifetime Total Cost of Ownership (LTV), factoring in fuel arbitrage, maintenance, and residual value, is a critical consumer consideration, we have purposefully excluded it as a regressor to maintain model integrity. In the Nepali context, LTV is virtually a proxy for the purchase price tax component due to stable energy costs and the nascent state of the secondary EV market. Including both triggers severe multicollinearity, muddying the 'signal' of the tax effect without adding incremental explanatory power. By isolating the tax variable, we ensure a higher-fidelity measurement of how fiscal policy directly moves the needle on market share

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Fitting logit model (standardized features + Ridge)...
      OLS Regression Results
=====
Dep. Variable:      logit_EV_Share    R-squared:      0.855
Model:              OLS              Adj. R-squared:  0.836
Method:             Least Squares     F-statistic:    141.0
Date:               Wed, 18 Mar 2026   Prob (F-statistic): 6.22e-32
Time:               10:05:11          Log-Likelihood: -62.732
No. Observations:   61               AIC:            141.5
Df Residuals:       53               BIC:            158.4
Df Model:           7
Covariance Type:    HC1
=====
               coef      std err      z      P>|z|      [0.025      0.975]
-----
const          -0.2247      0.198     -1.132     0.257     -0.614      0.164
Trend           0.9471      0.281      3.371     0.001      0.396      1.498
log_Tax_Ratio   -0.4827      0.071     -6.762     0.000     -0.623     -0.343
log_Price_Ratio -0.6237      0.243     -2.564     0.010     -1.100     -0.147
Q2              0.3757      0.251      1.498     0.134     -0.116      0.867
Q3              0.1132      0.262      0.431     0.666     -0.401      0.628
Q4              0.5271      0.311      1.697     0.090     -0.082      1.136
ImportBan       -0.0149      0.376     -0.040     0.968     -0.753      0.723
=====
Omnibus:          0.885      Durbin-Watson:    0.562
Prob(Omnibus):    0.642      Jarque-Bera (JB):  0.651
Skew:             -0.253      Prob(JB):          0.722
Kurtosis:         2.975      Cond. No.          6.35
=====

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Ridge Coefficients (alpha=1.0):
const: -0.169308
Trend: +0.906651
log_Tax_Ratio: -0.470488
log_Price_Ratio: -0.648370
Q2: +0.303972
Q3: +0.048239
Q4: +0.434233
ImportBan: -0.033036

```

Figure 3: Results Summary for Model 1

For years, analysts credited the 2021 "Paudel Reversal" and the 2022 import bans as the twin pillars of Nepal's electric vehicle surge. But a closer look at the data reveals a more nuanced story: Nepal's EV revolution was not only catalyzed by tax laws; it happened when the price tag changed. Our first econometric analysis (Model 1) highlights a critical truth: price is king. While the tax ratio carries a significant weight of -0.48, it is overshadowed by the coefficient of the price ratio at -0.62.

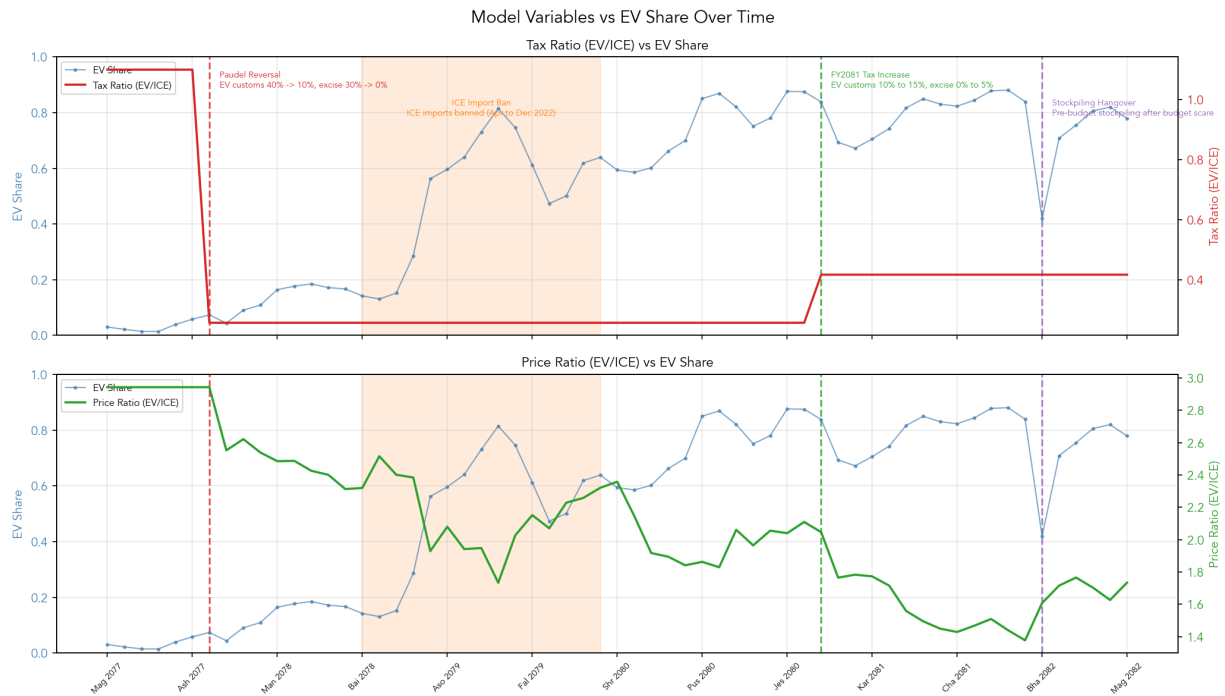


Figure 4: Graph of EV share over time, with policy variables outlined.

Figure 4 tells the story clearly. The blue line of EV market share remained relatively dormant through the early policy shifts of FY 2077 and 2078. The real "hockey stick" growth coincides almost perfectly with the sharp decline in the price ratio (the green line in the bottom chart).

The policy carrots were already in the room by 2022. What was missing was the product. The breakthrough news of 2023 to 2025 is effectively the story of the Chinese supply chain. By flooding the market with technologically advanced yet aggressively low-price vehicles, manufacturers like BYD, MG, and Neta did what the government couldn't: they crossed the affordability threshold.

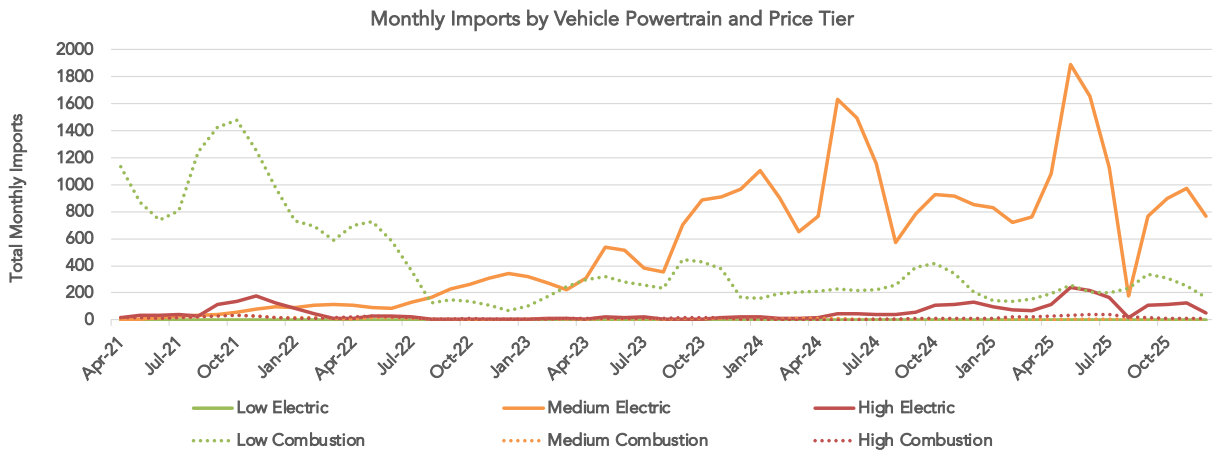


Figure 5: Breakdown of Monthly Imports

Figure 5 reinforces this price effect in the market. We categorized imported vehicles into three price tiers using percentiles of declared unit value: the bottom third (≤ 17 lakhs) is "low," the middle third (17–29 lakhs) is "medium," and the top third (> 29 lakhs) is "high." The trendlines suggest the clear substitution effect: consumers switched from low-tier ICE to EV, but a shift from low-tier, high-tax ICE vehicles to medium-tier EVs that faced lower taxes and delivered better features at roughly the equivalent showroom price.

Overall, the data suggests that Nepal's consumers are hyper-rational. They aren't switching to EVs solely because they are green, or even because they are significantly cheaper after taxes. They are switching because, for the same price they were paying before, they are getting a better product. To maintain EV adoption momentum, we must acknowledge that the current "Green Revolution" is largely subsidized by Chinese manufacturing efficiency. If those prices rise, the "policy story" will likely crumble, suggesting that in Nepal, the wallet drives the wheel.

Total Volumes – A Zero Sum Reality

In a healthy, growing economy, an "EV breakthrough" usually means the market is getting bigger: more people are buying cars for the first time. In Nepal, the room is sealed. In **Model 2 — Total Import Volume (Log-Linear Model)**, we model the log of total vehicle import quantities as a function of trend, average import cost, monthly

seasonality, and the import ban dummy. Continuous regressors are standardized and additionally estimated via ridge regression (L2 penalty, $\alpha = 1.0$):

$$\log(\text{TotalQty}_t) = \alpha + \beta_1 \text{Trend}_t + \beta_2 \text{AvgCost}_t + \gamma_{\text{month}(t)} + \delta \text{ImportBan}_t + \varepsilon_t$$

Here, γ denotes month fixed effects.

Note that instead of fiscal quarters, we use monthly dummies for this model. The EV share model uses quarterly dummies because the ratio of EVs to ICE vehicles shifts gradually with policy regimes and budget cycles. Total import volume, on the other hand, has sharp month-specific patterns driven by customs logistics and stocking behavior.

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OLS Regression Results

Dep. Variable:	log_Total_Qty_raw	R-squared:	0.561
Model:	OLS	Adj. R-squared:	0.428
Method:	Least Squares	F-statistic:	9.873
Date:	Wed, 18 Mar 2026	Prob (F-statistic):	1.27e-09
Time:	10:05:11	Log-Likelihood:	-33.874
No. Observations:	61	AIC:	97.75
Df Residuals:	46	BIC:	129.4
Df Model:	14		
Covariance Type:	HC1		

	coef	std err	z	P> z	[0.025	0.975]
const	6.3803	0.216	29.474	0.000	5.956	6.805
Trend	-0.0192	0.070	-0.275	0.783	-0.156	0.118
Avg_Cost	-0.0294	0.108	-0.273	0.785	-0.241	0.182
ImportBan	-0.8659	0.137	-6.297	0.000	-1.135	-0.596
M_Ashadh	-0.2791	0.358	-0.780	0.435	-0.980	0.422
M_Asoj	1.0991	0.256	4.294	0.000	0.597	1.601
M_Baishakh	0.8620	0.298	2.897	0.004	0.279	1.445
M_Bhadra	0.6889	0.255	2.699	0.007	0.189	1.189
M_Chaitra	0.4283	0.294	1.458	0.145	-0.147	1.004
M_Falgun	0.2631	0.295	0.891	0.373	-0.316	0.842
M_Jestha	1.1108	0.447	2.487	0.013	0.236	1.986
M_Kartik	0.5299	0.244	2.173	0.030	0.052	1.008
M_Magh	0.3665	0.298	1.232	0.218	-0.217	0.950
M_Mangsir	0.6074	0.222	2.734	0.006	0.172	1.043
M_Pus	0.7125	0.228	3.128	0.002	0.266	1.159

Omnibus:	5.692	Durbin-Watson:	1.683
Prob(Omnibus):	0.058	Jarque-Bera (JB):	5.249
Skew:	-0.463	Prob(JB):	0.0725
Kurtosis:	4.098	Cond. No.	14.1

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Volume Ridge Coefficients (alpha=1.0):

const:	+6.271592
Trend:	+0.014842
Avg_Cost:	-0.006956
ImportBan:	-0.600559
M_Ashadh:	-0.190132
M_Asoj:	+0.965671
M_Baishakh:	+0.761959
M_Bhadra:	+0.622908
M_Chaitra:	+0.447410
M_Falgun:	+0.316269
M_Jestha:	+0.967340
M_Kartik:	+0.489749
M_Magh:	+0.406775
M_Mangsir:	+0.551897
M_Pus:	+0.635904

Figure 6: Results Summary for Model 2

The results (trend coefficient = -0.02, with a low significance of 0.783) suggest that the total number of vehicles entering the country remains static. This means that for all the new brands, the shiny showrooms, and the "breakthrough" Chinese supply chains that saw the rapid rise of EVs in Nepal, the country is not buying more cars. It is simply buying *different* cars. Basically, market growth has hit a structural ceiling. We are not adding new drivers; we are simply swapping petrol engines for electric ones.

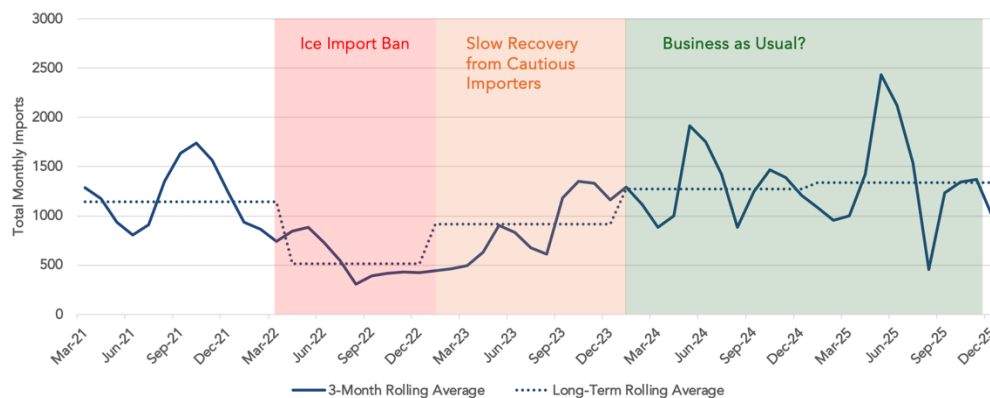


Figure 7: Monthly Trends in Total Consumer 4-Wheeler Imports

Figure 7 above reflects this reality. After the recovery from the ICE import ban, importers have failed to significantly exceed pre-April 2022 import volume. This indicates that the pool of customers with enough purchasing power seems to be capping out, and the vehicle retail business is now primarily relying on existing car owners trading in their old vehicles.

This is a classic zero-sum game. Because the market is saturated, the government cannot rely on volume growth to increase its tax revenue. Even as the total vehicle fleet grows, new import volume remains stagnant, and the government continues to give deep tax breaks to EVs, the national treasury faces a mathematical certainty of declining revenue. Policymakers must prepare for a significant revenue gap and consider alternative long-term fee structures (e.g., road-usage charges) to offset the loss of traditional import duties if the switching accelerates.

Modeling Limitations

While the regressions capture how EV share co-moves with relative taxes, relative import values, policy movements and broad time effects, several methodological caveats warrant disclosure.

First, the 3-month rolling smoothing used to stabilize monthly volatility induces serial correlation in the dependent variable; our HC1 standard errors correct for heteroskedasticity but not for this overlap. However, the smoothing is a deliberate trade-off: without it, Nepal's thin monthly import volumes produce sharp spikes at fiscal year boundaries and on customs clearance backlogs. The smoothed series better reflects the underlying demand conditions that the model is designed to capture, and the Durbin-Watson statistic is reported for transparency.

Second, the model cannot directly observe several drivers of consumer choice and market supply: vehicle availability and model mix, charging convenience and reliability, changes in quality or after-sales service, or informal pricing and invoicing practices that may affect declared import values. These omitted variables are a reality of working with flawed trade data in a developing economy. Critically, most of these factors — Chinese model launches, charging network expansion, after-sales improvements — trend monotonically upward over the sample period, meaning their effects are largely absorbed by the trend term rather than biasing the policy coefficients.

Finally, with 61 observations and 8 regressors, degrees of freedom are adequate but tight. This is an inherent constraint of Nepal's short EV history; there is simply no pre-2021 data of sufficient quality to extend the sample. Within this constraint, the model's adjusted R^2 of 0.836 and stable out-of-sample performance suggest that the specification captures the dominant sources of variation in the data.

Looking Out to the Future

To understand market sensitivities to tax changes, we simulate six scenarios over a 12-month horizon: the status quo and ICE tax reductions of 10% through 50% (in 10-percentage-point increments). Tax changes phase in over 3 months. For each scenario, we compute the predicted EV/ICE split using regression estimates and the resulting tax revenue, with low and high cases derived from the 95% confidence interval. The errors

across the two models are propagated to reflect a larger uncertainty than each of the models on its own.

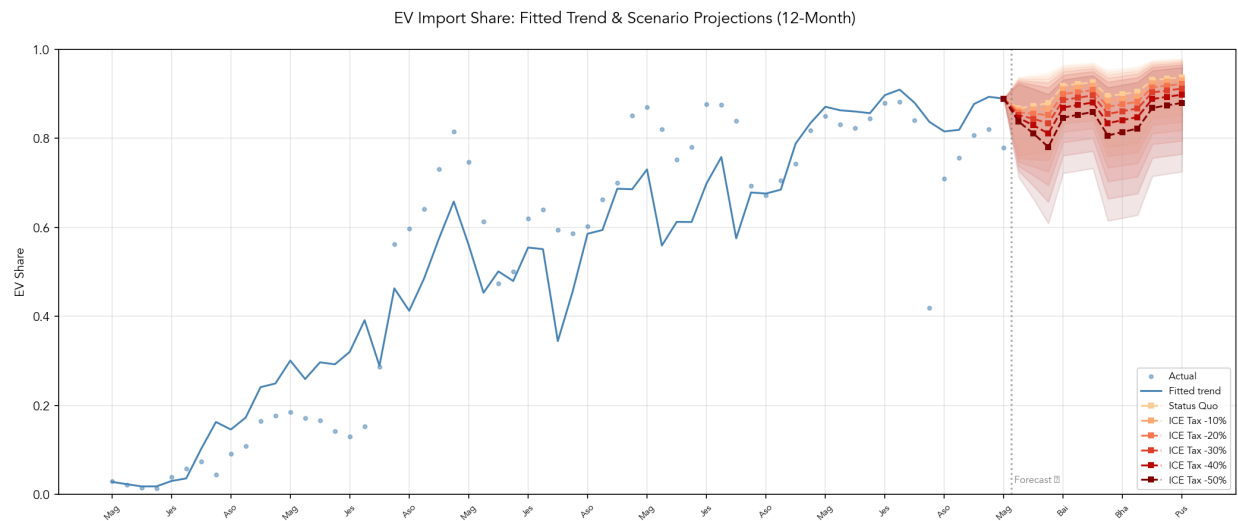


Figure 8: Fitted Trend of EV Share with Forecasts Under Various Scenarios

Figure 8, combined with the results from **Figure 2**, tell a fascinating story. A massive 50% cut to ICE taxes, an experiment that should normally cause panic to the government, only reduces vehicle tax revenue by 7 to 11% and overall tax revenue by up to negligible 0.7%. For moderate policy adjustments (10–20% tax cuts), the total tax revenue impact is statistically irrelevant.

This muted revenue impact stems from a composition effect. For decades, the Nepalese budget was built on the back of the petrol engine. Today, EVs represent over 70% of imports and are trending upward. Because combustion cars are now such a small part of the market, any reduction in tax revenue is minimized by the money coming from EVs. When ICE taxes are cut:

- Some marginal buyers switch from EVs to ICE vehicles
- ICE tax revenue per unit falls, but more ICE units are sold, partially offsetting the per-unit loss

Meanwhile, as Model 2 shows, a large portion of buyers are still priced out of the vehicle market altogether. These buyers would not be purchasing vehicles regardless of the powertrain and tax incentives, which speaks to our consumption power.

Our analysis points to a clear—and somewhat counterintuitive—implication for the Treasury: the government has room to take bolder steps. For years, automotive tax policy has been constrained by fears that lowering duties would “hollow out” revenue. But these models suggest that concern is overstated. Even under the “High Case” assumption of the 50% tax cut scenario, overall revenue remains remarkably resilient. That fiscal cushion creates space to redesign the tax system without risking a budget shock. Rather than relying on large upfront import duties, the government could shift toward a broader mix of recurring charges, as well as firming up internal revenue collection. This would make the “Sealed Room” (Model 2) more stable by aligning revenue with ongoing vehicle ownership, while still preserving a targeted green incentive for EVs.

The other crucial takeaway from our analysis, is that the binding constraint is the size of the market itself. Policies aimed at increasing household purchasing power, expanding vehicle financing, streamlining vehicle insurance or improving road infrastructure could expand the total addressable market. A 10% increase in total vehicle imports, for instance, would generate more additional revenue than any of the tax restructuring scenarios modeled here, regardless of the EV/ICE split.

If policy keeps chasing headlines, adoption will remain hostage to external price swings. But if policy redesigns the system around stable incentives and durable revenue, Nepal can lock in cleaner mobility and fiscal resilience at the same time.